

Spiral RoPE 🌀: Rotate Your Rotary Positional Embeddings in the 2D Plane

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Abstract

Rotary Position Embedding (RoPE) is the de facto positional encoding in large language models due to its ability to encode relative positions and serve as a foundation for length extrapolation techniques. When adapted to vision transformers, the standard axial formulation decomposes two-dimensional (2D) spatial positions into horizontal and vertical components, implicitly concentrating positional encoding on axis-aligned directions. We identify this property as a *directional bias* of the standard axial 2D RoPE, which biases the model away from oblique spatial relationships that naturally exist in natural images. To address this bias, we propose **Spiral RoPE**, a simple yet effective extension that enables multi-directional positional encoding by partitioning embedding channels into multiple groups associated with uniformly distributed directions. Each group is rotated according to the projection of the patch position onto its corresponding direction, allowing spatial relationships to be encoded beyond the horizontal and vertical axes. Across a wide range of vision tasks including classification, segmentation, and generation, Spiral RoPE consistently improves performance. Qualitative analysis of attention maps further show that Spiral RoPE exhibits more concentrated activations on semantically relevant objects and better respects local object boundaries, highlighting the importance of multi-directional positional encoding in vision transformers. Our code and model checkpoints are available at: https://github.com/haoyulianduzhuo-code/Spiral_RoPE.

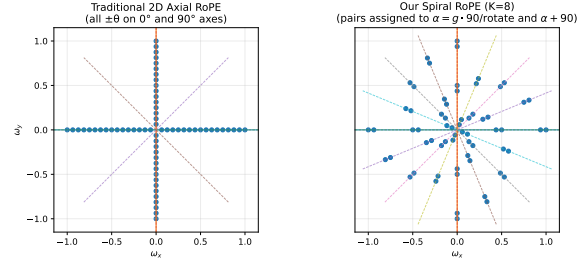


Figure 1. **Frequency support visualization** comparing Axial 2D RoPE (left) and Spiral RoPE (right) with eight rotation groups ($K = 8$). As shown, the Axial 2D RoPE places all frequencies on horizontal and vertical axes only, while our Spiral RoPE distributes frequencies across multiple directions in a spiral pattern, offering a broader directional coverage.

1. Introduction

Rotary Position Embedding (RoPE) (Su et al., 2024) has become a cornerstone of modern language transformers, powering large language models such as the LLaMA (Touvron et al., 2023a;b; Grattafiori et al., 2024) and Qwen (Bai et al., 2023; Yang et al., 2024; 2025b;a) series. By encoding positions through rotations on query and key vectors, RoPE provides a relative-position-aware inductive bias and is often more amenable to length extrapolation than absolute positional embeddings (APE). In contrast, standard Vision Transformers (ViTs) (Dosovitskiy et al., 2021) predominantly rely on APE which encodes fixed position vectors that are added to patch embeddings. While simple and effective, APE has well-documented limitations: it struggles to generalize to unseen resolutions and provides no explicit mechanism for encoding relative spatial relationships between image patches.

Recent work extends RoPE to vision tasks (Fang et al., 2024; Lu et al., 2024; 2023) with promising results on classification, dense prediction, and generation. The standard approach, which we term *Axial 2D RoPE*, decomposes 2D positions into horizontal and vertical components: half the embedding dimensions are rotated by the x -coordinate, the other half by the y -coordinate. However, this axial decomposition introduces a directional bias: its frequency support is concentrated on the coordinate axes, so diagonal or oblique spatial relationships are less directly encoded, although they may still be composed indirectly through the interaction of the x - and y -axis terms. This is a meaningful concern given

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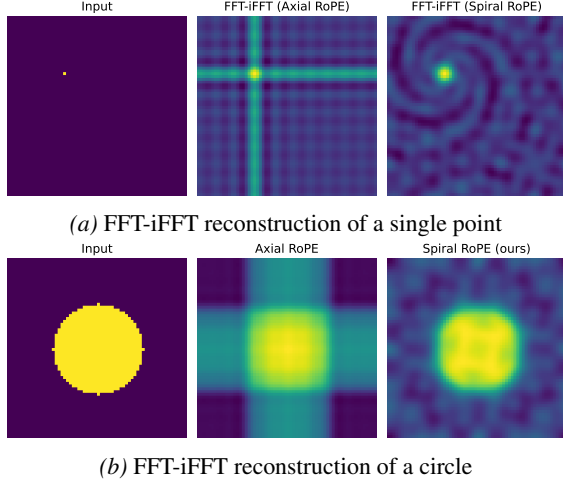


Figure 2. **Fourier reconstruction comparisons.** Given binary input images (left), we retain only frequencies representable by each RoPE method and reconstruct via inverse-FFT. Axial 2D RoPE (middle) produces artifacts along horizontal and vertical axes. Spiral RoPE achieves more faithful reconstruction due to broader directional coverage.

that natural images contain rich spatial structures in all orientations. To visualize this anisotropy, we perform a Fourier reconstruction experiment in Figure 2: retaining only the frequency components representable by each RoPE variant and reconstruct via inverse FFT (Fast Fourier Transform), Axial 2D RoPE produces visible artifacts along the coordinate axes, failing to faithfully reconstruct image structures such as circles.

In this work, we introduce **Spiral RoPE**, a new 2D rotary position embedding that extends directional coverage beyond the coordinate axes. Instead of splitting embeddings into just two axis-aligned groups, Spiral RoPE partitions them into K groups corresponding to K uniformly distributed directions in the 2D space. Each embedding group is rotated based on the patch position projected onto its assigned direction, enabling the model to explicitly encode positional relationships along diagonal and other oblique orientations. We design a grouped interleaved frequency assignment strategy that distributes the same number of distinct frequencies as Axial 2D RoPE across all K directions, ensuring no loss in multi-scale encoding capacity. As shown in Figure 1, the resulting frequency pattern forms a *spiral* in the 2D frequency plane, providing broader directional coverage.

Spiral RoPE introduces no extra computational overhead or learnable parameters compared to Axial 2D RoPE, yet yields consistent improvements across diverse vision tasks. On ImageNet-1k (Deng et al., 2009) classification, Spiral RoPE improves top-1 accuracy by +0.18% for DeiT-Large and +0.36% for DeiT-Base over Axial 2D RoPE. On ADE20k (Zhou et al., 2017) semantic segmentation, it achieves +1.22% mIoU for DeiT-Large and +0.53% mIoU for DeiT-Base. On class-conditional image generation with

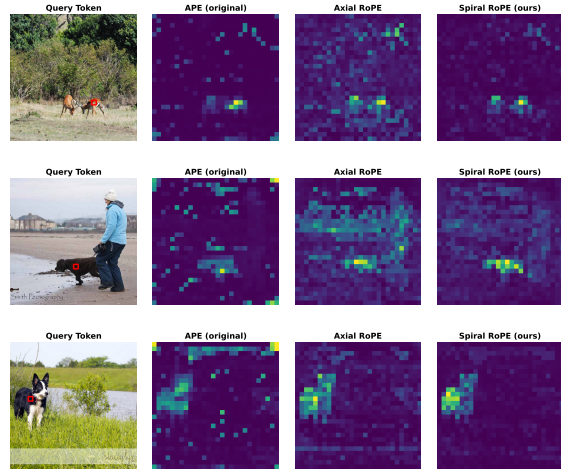


Figure 3. **Query-token attention visualization.** The query token (marked by a red box) is located on a foreground object. Across diverse scenes, Spiral RoPE consistently produces sharper and more localized attention around the queried region compared to APE and Axial RoPE, indicating improved spatial alignment.

Diffusion Transformers (DiT) (Peebles & Xie, 2023), it reduces FID by 0.7–1.1 points across various model sizes. We also observe that Spiral RoPE extrapolates better to unseen resolutions and unseen directions.

Qualitative analyses corroborate these gains. Figure 2 shows that Spiral RoPE enables more faithful Fourier-domain reconstruction than Axial 2D RoPE, avoiding characteristic axis-aligned artifacts. Consistent trends appear in the attention visualizations of Figure 3, where we visualize attention patterns from individual query tokens on the foreground object. Spiral RoPE produces sharper attention concentrated around the query location and semantically related regions, in contrast to the diffuse patterns of APE and the axis-biased activations of Axial RoPE. By broadening directional support through a simple geometric extension, Spiral RoPE provides a more flexible positional encoding for richer spatial interactions and improved spatial understanding in vision transformers.

2. Related Work

Positional Encoding in Transformers. Transformers (Vaswani et al., 2017) lack inherent positional awareness due to the permutation-equivariant nature of self-attention. To address this, the original Transformer used sinusoidal absolute position embeddings (APE), which add fixed position-dependent signals to input tokens. Learnable APE (Devlin et al., 2019) replaces fixed sinusoids with trainable vectors, offering greater flexibility. However, APE methods encode absolute positions, which limits their ability to generalize to sequence lengths unseen during training. Relative position encodings address this limitation by encoding the distance

between tokens rather than their absolute positions. Shaw et al. (Shaw et al., 2018) introduced learnable relative position biases added to attention logits. T5 (Raffel et al., 2020) further simplified this approach with a small number of learnable bias buckets. ALiBi (Press et al., 2022) proposed a non-learned linear bias that scales with distance, enabling length extrapolation in language models. When extending Transformers from 1D sequences to 2D visual data, vision Transformer (ViT) (Dosovitskiy et al., 2021) applies Transformers to images by dividing them into patches, using learnable APE. DeiT (Touvron et al., 2021) enabled efficient ViT training on ImageNet-1K, while Swin Transformer (Liu et al., 2021) introduced relative position bias within local windows. These developments highlight the importance of positional encoding design in vision models for handling varying resolutions and capturing spatial relationships.

Rotary Position Embedding. Rotary Position Embedding (RoPE) (Su et al., 2024) introduces a fundamentally different approach to position encoding by applying rotation operations directly to query and key vectors. Unlike additive position encodings, RoPE multiplies the query and key representations by rotation matrices whose angles are determined by the position index and a set of base frequencies. This design ensures that the dot product between query and key depends only on their relative positions, inherently encoding relative positional information. RoPE has become the de facto position encoding for modern large language models including LLaMA (Touvron et al., 2023a) and Qwen (Yang et al., 2025a). While vanilla RoPE itself does not reliably extrapolate to much longer sequences, it has served as the foundation for a number of follow-up works that address specific limitations of the formulation: position interpolation (Chen et al., 2023) rescales positional indices at inference time and YaRN (Peng et al., 2024) refines this strategy through frequency-aware scaling, both targeting long-context extrapolation. More recently, PoPE (Gopalakrishnan et al., 2025) analyses the entanglement between content and position in RoPE and proposes a polar-coordinate variant to decouple the two; and CoPE (Li et al., 2026) leverages frequency clipping for long-context adaptation.

2D RoPE for Vision. Extending RoPE to 2D vision data requires careful design choices. The straightforward approach, which we term axial 2D RoPE, splits the embedding dimension into two halves and applies 1D RoPE independently along the x and y axes. This 2D RoPE has been adopted in many vision tasks (Wang et al., 2026), such as SAM 2 (Ravi et al., 2025) for image segmentation and Qwen-Image (Wu et al., 2025) for image generation. M-RoPE (Multimodal RoPE) (Wang et al., 2024) decomposes image indices into height, width, and temporal components to align visual patches with 1D text sequences. However, these axial approaches encode only horizontal and verti-

cal relationships, without explicit awareness of diagonal or other directional spatial patterns. The most closely related prior work is RoPE-Mixed (Heo et al., 2024), which treats the frequencies for both spatial axes as learnable parameters, allowing the network to adaptively “mix” axial signals to capture diagonal information. In multimodal settings, several works generalize RoPE to jointly handle tokens from different modalities. For instance, Circle-RoPE (Wang et al., 2025) project image token indices onto a ring that is orthogonal to the linear axis of text token indices to eliminate spurious cross-modal biases.

3. Method

3.1. Preliminaries: Rotary Position Embedding

3.1.1. 1D RoPE

Rotary Position Embedding (Su et al., 2024) encodes positional information through rotation operations. Given a d -dimensional query or key vector $\mathbf{x} \in \mathbb{R}^d$ at position m , RoPE applies a block-diagonal rotation matrix:

$$\text{RoPE}(\mathbf{x}, m) = \mathbf{R}_m \mathbf{x} \quad (1)$$

where $\mathbf{R}_m$ is a block-diagonal matrix composed of $d/2$ 2D rotation matrices:

$$\mathbf{R}_m = \begin{pmatrix} \mathbf{R}_m^{(0)} & & \\ & \ddots & \\ & & \mathbf{R}_m^{(d/2-1)} \end{pmatrix} \quad (2)$$

with each 2×2 block being:

$$\mathbf{R}_m^{(t)} = \begin{pmatrix} \cos(m\theta_t) & -\sin(m\theta_t) \\ \sin(m\theta_t) & \cos(m\theta_t) \end{pmatrix} \quad (3)$$

The frequency θ_t is typically set as $\theta_t = \theta_{\text{base}}^{-t/(d/2)}$ for $t = 0, 1, \dots, d/2 - 1$, with θ_{base} commonly set to 10,000, giving a geometric frequency progression that encodes positional dependencies at multiple scales. A key property of RoPE is that the dot product between rotated vectors depends only on their relative position:

$$\langle \text{RoPE}(\mathbf{q}, m), \text{RoPE}(\mathbf{k}, n) \rangle = \mathbf{q}^\top \mathbf{R}_{m-n} \mathbf{k}, \quad (4)$$

so attention scores implicitly encode relative positional information when calculating the attention matrix in the self-attention mechanism.

3.1.2. AXIAL 2D ROPE

RoPE was originally designed for 1D sequences. For 2D images, the standard approach is to extend RoPE by decomposing the spatial position into x and y coordinates. Given

a patch at position (p_x, p_y) , the d -dimensional embedding $\mathbf{x} \in \mathbb{R}^d$ is split into two halves:

$$\mathbf{x}^{(x)} = (x_0, x_1, \dots, x_{d/2-1}) \quad (5)$$

$$\mathbf{x}^{(y)} = (x_{d/2}, x_{d/2+1}, \dots, x_{d-1}) \quad (6)$$

The first half is rotated using the x-coordinate, and the second half using the y-coordinate:

$$\text{RoPE-2D}(\mathbf{x}, p_x, p_y) = \begin{pmatrix} \mathbf{R}_{p_x} \mathbf{x}^{(x)} \\ \mathbf{R}_{p_y} \mathbf{x}^{(y)} \end{pmatrix} \quad (7)$$

The rotation matrices $\mathbf{R}_{p_x}$ and $\mathbf{R}_{p_y}$ are typically constructed using the same set of frequencies. Each sub-embedding $\mathbf{x}^{(x)}$ and $\mathbf{x}^{(y)}$ has dimensionality $d/2$ and thus can only accommodate $d/4$ distinct rotary frequencies, i.e., half of those available in the original 1D RoPE.

Despite the simplicity of transferring 1D RoPE to 2D inputs, this axial design only encodes positional relationships along the axis-aligned directions, leading to a clear directional bias. The x-coordinate and y-coordinate are processed independently. Consider patches at positions $\mathbf{p}_1 = (0, 0)$ and $\mathbf{p}_2 = (1, 1)$, which are diagonally adjacent. With conventional 2D RoPE, the relative position encoding in the x-dimensions reflects $\Delta x = 1$ and in the y-dimensions reflects $\Delta y = 1$, but there is no direct encoding of the diagonal relationship.

3.2. Spiral RoPE

To address this directional bias, we extend 2D RoPE with additional *spatial rotations* on the 2D panel. More generally, we encode positions along multiple uniformly distributed directions, providing a more complete description of spatial relationships.

Formally, let K be the number of directions we want to encode. We uniformly distribute K directions in the angular range $[0, \pi)$:

$$\phi_k = \frac{k\pi}{K}, \quad k = 0, 1, \dots, K-1 \quad (8)$$

Note that we only need to cover $[0, \pi)$ rather than $[0, 2\pi)$ because directions ϕ and $\phi + \pi$ are parallel, and RoPE naturally handles both positive and negative positions through its rotation mechanism.

For each direction ϕ_k , we define a unit direction vector:

$$\mathbf{u}_k = (\cos \phi_k, \sin \phi_k) \quad (9)$$

Given a patch at 2D position $\mathbf{p} = (p_x, p_y)$, we compute its projected position along direction k as:

$$t_k(\mathbf{p}) = \mathbf{p} \cdot \mathbf{u}_k = p_x \cos \phi_k + p_y \sin \phi_k \quad (10)$$

We partition the d -dimensional embedding into K equal groups, each with d/K dimensions. The k -th group, denoted $\mathbf{x}^{(k)}$, is rotated using the projected position t_k :

$$\text{Spiral RoPE}(\mathbf{x}, \mathbf{p}) = \begin{pmatrix} \mathbf{R}_{t_0(\mathbf{p})}^{(0)} \mathbf{x}^{(0)} \\ \mathbf{R}_{t_1(\mathbf{p})}^{(1)} \mathbf{x}^{(1)} \\ \vdots \\ \mathbf{R}_{t_{K-1}(\mathbf{p})}^{(K-1)} \mathbf{x}^{(K-1)} \end{pmatrix} \quad (11)$$

where $\mathbf{R}_t^{(k)}$ denotes the rotation matrix for direction k with its specific frequency assignment.

A critical design choice in Spiral RoPE is how to assign rotation frequencies θ_i to different directions. A naive approach would independently assign $d/(2K)$ frequencies to each direction. However, this limits the number of distinct frequencies as K increases, reducing the model's ability to encode positions at multiple scales.

To maximize frequency diversity, we adopt an interleaved frequency assignment strategy. We first compute a pool of $d/4$ base frequencies using the same formula as axial 2D RoPE:

$$\theta_t = \theta_{\text{base}}^{-t/(d/4)}, \quad t = 0, 1, \dots, d/4 - 1 \quad (12)$$

This gives us the same number of distinct frequencies as the conventional 2D RoPE baseline.

To distribute these frequencies across directions, we adopt a grouped interleaved assignment strategy. We first group adjacent frequencies into pairs: $(\theta_0, \theta_1), (\theta_2, \theta_3), \dots, (\theta_{d/4-2}, \theta_{d/4-1})$. We then assign these pairs to direction groups in a round-robin fashion. Specifically, we pair directions that are 90° apart (i.e., perpendicular), since they encode orthogonal spatial relationships. For direction pair $(k, k + K/2)$ where $k < K/2$, we assign frequency pairs:

$$\Theta^{(k)} = \Theta^{(k+K/2)} = \{\theta_{2k}, \theta_{2k+1}, \theta_{2k+K}, \theta_{2k+K+1}, \dots\} \quad (13)$$

For example, with $K = 4$ directions ($0^\circ, 45^\circ, 90^\circ, 135^\circ$) and $d = 32$, the frequency assignment is:

- $\theta_0, \theta_1, \theta_4, \theta_5$ are assigned to directions 0° and 90°
- $\theta_2, \theta_3, \theta_6, \theta_7$ are assigned to directions 45° and 135°

Figure 1 visualizes the frequency assignment by plotting each frequency $\pm\theta_t$ as a point in the 2D frequency plane along the direction it encodes. For traditional 2D axial RoPE, all frequencies lie on the horizontal (0°) and vertical (90°) axes, corresponding to frequency vectors $(\pm\theta_t, 0)$ and $(0, \pm\theta_t)$. In contrast, Spiral RoPE with $K = 8$ distributes

frequencies across $K/2 = 4$ pairs of perpendicular directions: $0^\circ/90^\circ$, $22.5^\circ/112.5^\circ$, $45^\circ/135^\circ$, and $67.5^\circ/157.5^\circ$, which forms a *spiral* pattern. This visualization illustrates how Spiral RoPE expands directional coverage beyond the coordinate axes while maintaining the same total number of frequencies.

This grouped interleaved assignment ensures that: (1) we utilize all $d/4$ distinct frequencies, matching the capacity of axial 2D RoPE; (2) perpendicular direction pairs share the same frequency set, analogous to how axial 2D RoPE uses the same frequencies for both x and y axes; and (3) each direction receives a mix of low and high frequencies from adjacent pairs, enabling multi-scale positional encoding within each direction.

Spiral RoPE preserves RoPE’s relative-position property: since $t_k(\mathbf{p}) - t_k(\mathbf{p}') = t_k(\mathbf{p} - \mathbf{p}')$, the attention contribution depends only on the relative position $\mathbf{p} - \mathbf{p}'$. By uniformly distributing directional encodings, Spiral RoPE provides more isotropic spatial awareness than axial 2D RoPE, and can be implemented efficiently by precomputing rotation matrices, with negligible overhead.

To further illustrate the representation power of Spiral RoPE, we perform a 2D Fourier analysis with $d = 1024$ ($d/4 = 256$ distinct frequencies). Given a binary image, we first apply 2D FFT to obtain its frequency-domain representation. We then retain only the frequency components that can be represented by each RoPE variant: for axial 2D RoPE, we keep frequencies along the horizontal and vertical axes ($\pm\theta_t, 0$) and $(0, \pm\theta_t)$; for Spiral RoPE, we keep frequencies along all $K = 8$ directions as illustrated in Figure 1. All other frequency components are masked to zero. Finally, we apply 2D iFFT to reconstruct the image and compare the result with the original input. Figure 2 shows reconstructions for a single point and a circle on a 64×64 grid: axial 2D RoPE exhibits visible artifacts along the coordinate axes, whereas Spiral RoPE produces reconstructions closer to the original images, demonstrating broader directional coverage.

4. Experiments

We conduct comprehensive experiments to validate the effectiveness of Spiral RoPE across a wide range of vision tasks. We begin with a controlled synthetic spatial reasoning task that directly probes whether the additional directional groups in Spiral RoPE improve off-axis spatial reasoning, and then move on to large-scale evaluations: image classification on ImageNet-1k, semantic segmentation on ADE20k, and class-conditional image generation on ImageNet. Our experiments demonstrate that Spiral RoPE consistently improves performance over baselines and prior 2D RoPE methods.

4.1. Synthetic Spatial Reasoning

To directly test whether Spiral RoPE’s broader directional support translates into improved off-axis spatial reasoning, we design two controlled synthetic tasks on 224×224 images of coloured patches on a neutral background, training a DeiT-Tiny (Touvron et al., 2021) from scratch and comparing Spiral RoPE against standard Axial 2D RoPE under identical settings (full details in Appendix B.1). The *direction* task is an 8-way classification of which angular bin (E, NE, N, NW, W, SW, S, SE) the blue target patch lies in relative to the red reference patch (with six random-colour distractors). Figure 4 show representative samples from the direction task. The *distance* task is a 4-way classification of the patch separation distance (Near, Mid-near, Mid-far, Far) under a directional distributional shift: training contains only axis-aligned (E, N, W, S) placements, whereas the test set additionally includes diagonal (NE, NW, SW, SE) placements as out-of-distribution (OOD) examples at the same distance bins. Figure 5 show representative samples from the distance task. Table 1 reports the results. On the direction task, Spiral RoPE reduces the overall error from 0.95% to 0.11%, with the largest gain on diagonal cases ($1.10\% \rightarrow 0.16\%$). On the distance task, both methods nearly saturate on the in-distribution axis-aligned subset, but on the OOD diagonal subset Axial RoPE incurs 70.38% error while Spiral RoPE retains only 41.72%, a 28.66-point reduction. Together, these results provide direct, controlled evidence that Spiral RoPE’s broader directional support improves off-axis spatial reasoning.

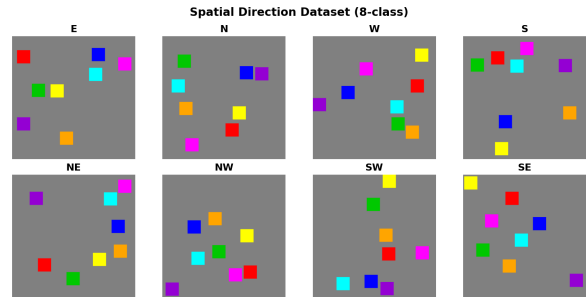


Figure 4. Sample images from the relative direction classification dataset. The model classifies which of 8 directions the blue target patch lies relative to the red reference patch, with six random-colour distractor patches added.

4.2. Image Classification

Setup. We evaluate Spiral RoPE on ImageNet-1k (Deng et al., 2009) classification using DeiT (Touvron et al., 2021) as the base architecture. We train DeiT-Small, DeiT-Base and DeiT-Large models with and without Spiral RoPE on three target resolutions: 224×224 , 384×384 , and 448×448 . Following (Heo et al., 2024), we combine Spiral RoPE with the original absolute positional embedding (APE), as this

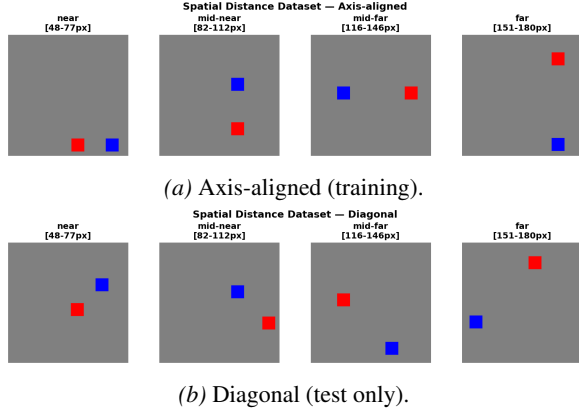


Figure 5. Sample images from the distance classification dataset. The model is trained only on axis-aligned placements (a) and evaluated on both axis-aligned and diagonal placements (b).

Table 1. *Synthetic spatial reasoning error rates (%)*. Left: direction classification, errors split by ground-truth direction. Right: distance classification with training restricted to axis-aligned placements, evaluated on axis-aligned (ID) and diagonal (OOD) test placements.

METHOD	DIRECTION TASK			DISTANCE TASK		
	AXIS	DIAG	ALL	ID	OOD	ALL
AXIAL ROPE	0.80	1.10	0.95	0.90	70.38	35.64
SPIRAL ROPE (OURS)	0.06	0.16	0.11	0.42	41.72	21.07

combination has been shown to be effective. We use $K = 16$ directions for Spiral RoPE with all frequencies θ_t scaled by a factor of 1.5 (i.e., $\theta_t = 1.5\theta_{\text{base}}^{-t/(d/4)}$). Models are pre-trained at 224×224 resolution for 300 epochs (Small and Base) or 200 epochs (Large), then fine-tuned at the target resolution for 20 epochs. We compare against the standard DeiT baseline (APE) and Axial RoPE. We also compare with RoPE-Mixed (Heo et al., 2024), which uses learnable per-head frequencies, in contrast to our method that uses pre-specified directions and frequencies. For evaluation, we report Top-1 accuracy on the validation set using the final checkpoint with exponential moving average (EMA) weights and full crop (crop ratio = 1.0). Additional training details are provided in Appendix B.

Results. Table 2 presents the classification results across different resolutions. Spiral RoPE consistently outperforms the baselines across various configurations. These results demonstrate that the multi-directional encoding in Spiral RoPE provides richer positional information that benefits image classification, with larger improvements observed at higher resolutions. We additionally verify the statistical reliability of these gains via 3-seed runs of DeiT-Large in Appendix A: standard errors fall in 0.03–0.06, well below the gaps between methods.

Table 2. *ImageNet-1k classification accuracy (%)*. Models are pre-trained at 224×224 and fine-tuned at each target resolution. Spiral RoPE consistently outperforms baselines across all configurations.

MODEL	METHOD	224	384	448
DeiT-S	APE	79.11	81.88	82.24
	AXIAL ROPE	80.06	82.53	82.98
	ROPE-MIXED	80.48	82.64	82.99
	SPIRAL ROPE (OURS)	80.39	83.04	83.15
DeiT-B	APE	82.36	84.16	84.29
	AXIAL ROPE	83.21	84.91	85.09
	ROPE-MIXED	83.15	84.73	84.84
	SPIRAL ROPE (OURS)	83.39	85.04	85.19
DeiT-L	APE	83.24	84.92	85.13
	AXIAL ROPE	83.61	85.19	85.42
	ROPE-MIXED	83.51	85.14	85.40
	SPIRAL ROPE (OURS)	83.97	85.48	85.67

Extrapolation Robustness. We evaluate robustness to unseen input resolutions using models trained at 224×224 and tested across resolutions from 144×144 to 512×512 . Other evaluation settings are the same as before. For APE, position vectors are bicubically interpolated; for RoPE variants, position indices extend naturally to the new grid. As shown in Figure 6, all three RoPE variants outperform APE across resolutions and model sizes. The RoPE variants perform comparably near the training resolution, but at higher resolutions (384–512), Spiral RoPE exhibits a clear advantage over both Axial RoPE and RoPE-Mixed, suggesting that our pre-specified directions and fixed sinusoidal frequencies extrapolate better than the axis-restricted or learned alternatives.

Rotation Robustness. We further test whether Spiral RoPE confers a stronger inductive bias for off-axis spatial structures by evaluating zero-shot robustness to global image rotations: using DeiT-Base models trained at 224×224 (no rotation augmentation), we rotate every image in the ImageNet-1k validation set by angles between 0° and 180° and re-evaluate top-1 accuracy. As shown in Figure 7, Spiral RoPE consistently outperforms Axial RoPE at every angle, and crucially the gap is largest at the maximally off-axis angles (+2.65 at 45° , +3.18 at 135°), while it nearly vanishes at angles that keep image content axis-aligned (+0.18 at 0° , +0.55 at 90° , +0.07 at 180°). This pattern matches our analysis in Sec. 3: Axial RoPE is most disadvantaged precisely when the dominant image structure no longer aligns with the coordinate axes, whereas Spiral RoPE’s broader directional coverage provides robust encoding across all orientations.

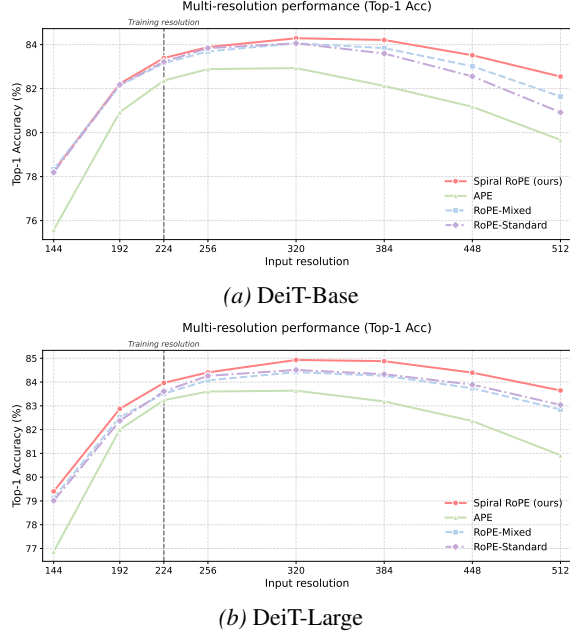


Figure 6. **Multi-resolution performance on ImageNet-1k.** Models are trained at 224×224 and evaluated across resolutions from 144×144 to 512×512 . Spiral RoPE demonstrates superior robustness to resolution changes compared to APE, Axial RoPE, and RoPE-Mixed baselines across both model sizes.

4.3. Semantic Segmentation

Setup. We evaluate Spiral RoPE on semantic segmentation using the ADE20k dataset (Zhou et al., 2017; 2019), which contains 20,000 training images and 2,000 validation images across 150 semantic categories. We use UperNet (Xiao et al., 2018) as the segmentation framework with ViT training recipe, and we initialize the backbone with the ImageNet pre-trained weights at 224×224 . The segmentation models are trained at 512×512 resolution for 160k iterations with a batch size of 16. For evaluation, we perform single-scale testing on the validation set and report mean Intersection over Union (mIoU), mean class accuracy (mAcc), and overall pixel accuracy (aAcc). Additional training details are provided in Appendix B.

Results. As shown in Table 3, Spiral RoPE achieves the best results across all backbones and metrics. The gains are more pronounced than in classification, suggesting that for dense prediction tasks such as semantic segmentation where reasoning is more complex and higher input resolutions are required, a well-designed relative position embedding can more effectively facilitate the modeling of spatial dependencies, demonstrating that Spiral RoPE consistently delivers larger benefits on more challenging tasks. The 3-seed DeiT-Large analysis in Appendix A confirms that the mIoU gap is an order of magnitude larger than the standard error.

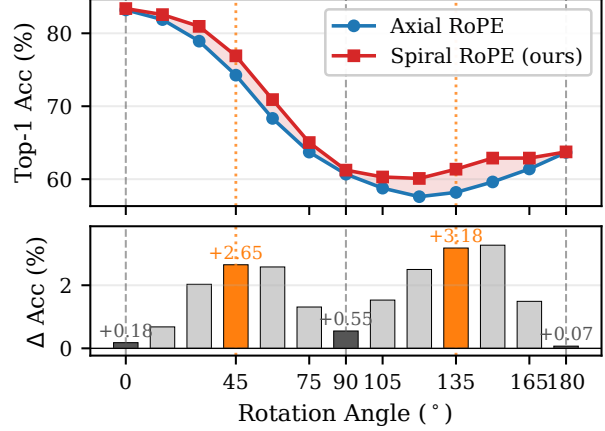


Figure 7. **Zero-shot rotation robustness on ImageNet-1k classification (DeiT-Base).** Both models are trained at 224×224 using the same configuration as Sec. 4.2 (no rotation augmentation). Validation images are globally rotated by the listed angle. *Top:* top-1 accuracy of Axial RoPE and Spiral RoPE; the gap (shaded) widens at off-axis rotations and narrows at axis-aligned ones. *Bottom:* per-angle improvement of Spiral RoPE over Axial RoPE; dark bars mark axis-aligned rotations (0° , 90° , 180°) and orange bars highlight the maximally off-axis rotations (45° , 135°).

4.4. Image Generation

Setup. We evaluate Spiral RoPE on class-conditional image generation using Diffusion Transformers (DiT) (Peebles & Xie, 2023), training DiT-S/B/L/XL with patch size 2 on ImageNet 256×256 (16×16 tokens after patchification). We use $K = 8$ directions for S/B/L and $K = 6$ for XL, training for 400k steps with the original DiT (Peebles & Xie, 2023) recipe. For evaluation, we generate 50k samples using DDPM with 250 steps and classifier-free guidance scale 1.0, and report Fréchet Inception Distance (FID) (Heusel et al., 2017) against the ImageNet validation set. Additional details are in Appendix B.

Results. As shown in Table 4, Spiral RoPE achieves the best FID across all DiT model sizes. The results suggest that Spiral RoPE is most effective when the sequence length is sufficiently large for the multi-directional encoding to capture meaningful spatial relationships.

Extended Training. To further demonstrate the effectiveness of Spiral RoPE when scaled to longer training, we conduct experiments following the setup of SiT (Ma et al., 2024). Specifically, we train the XL/2 models for 7 million steps on ImageNet 256×256 and evaluate using classifier-free guidance with scale 1.5. Table 5 compares our Spiral RoPE against the original DiT baseline and SiT. Our Spiral RoPE achieves an FID of 1.74, outperforming both DiT (2.27) and SiT (2.06). Figure 8 shows qualitative samples generated by our model.

Table 3. *ADE20k semantic segmentation results.* All models are pretrained on ImageNet-1k and then finetuned with an UperNet segmentation head for 160k iterations at 512×512 resolution.

BACKBONE	METHOD	mIOU	mACC	AACC
DEiT-S	APE	43.72	54.33	81.07
	AXIAL ROPE	44.98	55.13	81.32
	ROPE-MIXED	45.05	55.16	81.40
	SPIRAL ROPE	45.44	56.33	81.61
DEiT-B	APE	46.89	57.28	82.30
	AXIAL ROPE	47.58	58.29	82.56
	ROPE-MIXED	48.00	58.46	82.56
	SPIRAL ROPE	48.11	59.07	82.76
DEiT-L	APE	46.91	57.05	82.36
	AXIAL ROPE	47.90	58.87	83.12
	ROPE-MIXED	48.55	59.28	83.38
	SPIRAL ROPE	49.12	60.11	83.40



Figure 8. *Generated samples* from our DiT-XL/2 model with Spiral RoPE trained for 7M steps on ImageNet 256×256 . Images are generated using classifier-free guidance with scale 4.0.

5. Analysis

In this section, we provide further analysis of Spiral RoPE, including visualization of attention maps and ablation studies on key hyperparameters.

5.1. Visualization of Attention Maps

To qualitatively understand the differences between positional encoding methods, we visualize attention maps from DeiT-Base models fine-tuned at 448×448 resolution. We compare three variants: the original APE baseline, APE with axial 2D RoPE, and APE with Spiral RoPE ($K = 16$).

Class-Token Attention. We first visualize the class-token attention by averaging the attention weights over all heads in the last transformer layer. Figure 9 shows representative examples from the ImageNet validation set. A clear progres-

Table 4. *Class-conditional image generation* on ImageNet 256×256 using DiT. We compare models trained for 400k steps with the standard training hyperparameters of DiT. We generate 50k samples using DDPM sampling with 250 steps and classifier-free guidance scale of 1.0. Baseline results with supermarks * are taken from U-DiTs (Tian et al., 2024).

MODEL	METHOD	FID↓	SFID↓	IS↑	PREC.↑	REC.↑
DiT-S/2	APE*	67.40	11.93	20.44	0.368	0.559
	AXIAL ROPE	64.28	11.94	21.05	0.375	0.564
	ROPE-MIXED	63.92	11.92	21.80	0.380	0.574
	SPIRAL ROPE	63.33	11.91	21.96	0.386	0.574
DiT-B/2	APE*	42.84	8.24	33.66	0.491	0.629
	AXIAL ROPE	39.88	8.24	36.36	0.511	0.629
	ROPE-MIXED	39.90	8.24	36.78	0.508	0.628
	SPIRAL ROPE	37.74	8.23	38.31	0.520	0.630
DiT-L/2	APE*	23.27	6.35	59.63	0.611	0.635
	AXIAL ROPE	20.06	6.38	64.19	0.623	0.633
	ROPE-MIXED	20.73	6.37	63.77	0.624	0.633
	SPIRAL ROPE	19.02	6.36	69.49	0.638	0.632
DiT-XL/2	APE*	20.05	6.25	66.74	0.632	0.629
	AXIAL ROPE	16.25	6.01	70.58	0.649	0.627
	ROPE-MIXED	15.97	6.14	72.43	0.655	0.625
	SPIRAL ROPE	15.55	5.87	79.67	0.662	0.631

Table 5. *Extended training comparison* on ImageNet 256×256 . The Spiral RoPE model uses DiT-XL/2 architecture trained for 7M steps. We use classifier-free guidance with scale 1.5. Baseline results are from SiT (Ma et al., 2024).

METHOD	FID↓	SFID↓	IS↑	PREC.↑	REC.↑
DiT	2.27	4.60	278.24	0.83	0.57
SiT	2.06	4.49	277.50	0.83	0.59
SPIRAL ROPE (OURS)	1.74	4.49	279.64	0.83	0.60

sion is visible across the three methods. The APE baseline produces diffuse attention patterns, with substantial activation spread across both foreground objects and background regions. Axial RoPE improves over APE by allocating stronger attention to semantically relevant regions; however, its attention maps still exhibit noticeable residual activations in surrounding background areas and irrelevant objects. In contrast, Spiral RoPE yields the most object-centric and spatially coherent attention patterns, with concentrated responses on the foreground objects and effective suppression of background activations. This advantage is particularly pronounced in multi-instance scenes, where Spiral RoPE sharply attends to multiple relevant objects, while Axial RoPE still exhibits axis-aligned artifacts and attention leakage into nearby regions.

Query-Token Attention. To further examine how positional encoding influences local spatial relationships, we visualize attention patterns originating from individual patch tokens. Figure 3 shows representative examples in which the query token (marked by a red box) is located on a foreground object. Across all examples, APE and Axial RoPE

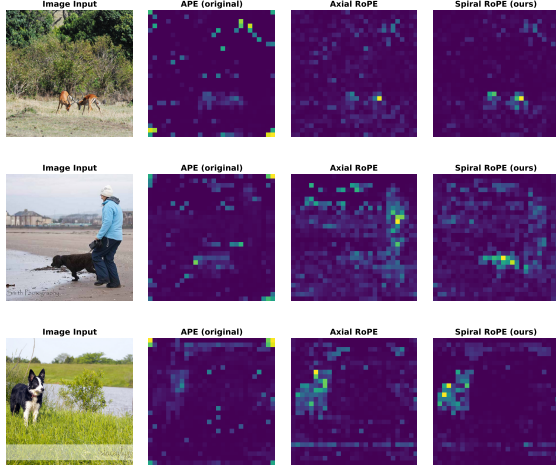


Figure 9. *Class-token attention visualization.* Each row shows the input image followed by attention maps from APE, Axial 2D RoPE, and Spiral RoPE. Attention is averaged over all heads in the last layer. Spiral RoPE produces more object-centric attention with cleaner background suppression compared to both counterparts.

produce attention maps that are broadly distributed across the image, with substantial activation on background regions and unrelated objects. In contrast, Spiral RoPE consistently produces markedly sharper and more spatially coherent attention patterns, with strong responses concentrated around the query location and semantically related regions. This suggests that Spiral RoPE facilitates more precise spatial alignment between query and key tokens, enabling attention to better respect local object boundaries and spatial proximity.

5.2. Ablation Studies

We ablate key hyperparameters of Spiral RoPE on ImageNet-1k classification with DeiT-Base at 224×224 , using the same training recipe and evaluation settings as Sec. 4.2.

Number of Directions K . We first ablate the number of directions K in Spiral RoPE with the frequency scaling factor fixed at 1.5. As shown in Table 6, Spiral RoPE consistently outperforms the axial RoPE baseline (83.31%) across all values of K , with $K = 16$ achieving the best performance (83.39%) and further increasing K to 32 slightly degrading performance (83.32%). This reflects a trade-off intrinsic to Spiral RoPE: at fixed head dimension d , our interleaved scheme assigns $d/(2K)$ sin-cos pairs per direction, so larger K broadens angular coverage at the cost of multi-scale (frequency) capacity within each direction. We therefore view K as a tunable hyperparameter: a larger head dimension can afford more frequencies per direction and may support a larger optimal K , and the best value may also vary across architectures and application scenarios.

Frequency Scaling Factor. We also ablate the scaling factor applied to all base frequencies θ_t , with $K = 16$. As shown in Table 7, accuracy peaks at scale 1.5 (83.39%); larger values (1.75, 2.0) degrade performance, likely because excessively high frequencies make rotations vary too rapidly for stable learning. We therefore use 1.5 as the default setting.

Table 6. Ablation on the number of directions K .

K	ACC (%)
2 (AXIAL ROPE)	83.21
4	83.31
8	83.34
16	83.39
32	83.32

Table 7. Ablation on the frequency scaling factor.

SCALE	ACC (%)
1.00	83.33
1.25	83.13
1.50	83.39
1.75	83.34
2.00	83.23

6. Conclusion

We have presented Spiral RoPE, a simple yet effective extension of rotary position embedding for vision transformers that addresses the directional bias of Axial 2D RoPE by distributing positional encoding across K uniformly spaced directions. Across image classification, semantic segmentation, and image generation, Spiral RoPE delivers consistent improvements with particularly strong resolution extrapolation, suggesting that directional diversity in positional encoding is critical for modeling 2D spatial relationships.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Statistical Reliability of Large-scale Results

To characterize the variance of our large-scale results, we re-run the DeiT-Large experiments on ImageNet-1k classification and ADE20k semantic segmentation with three independent seeds and report the mean $\pm$ standard error in Table 8. The standard errors are small (all within 0.03–0.06) and substantially smaller than the gaps between methods, indicating that the ordering $\text{APE} < \text{Axial RoPE} < \text{RoPE-Mixed} < \text{Spiral RoPE}$ is statistically reliable. In particular, the Spiral RoPE gain over the next-best baseline (RoPE-Mixed) is +0.44 accuracy on ImageNet-1k and +0.61 mIoU on ADE20k, both an order of magnitude larger than the corresponding standard errors.

Table 8. **DeiT-Large results with standard errors.** ImageNet-1k classification top-1 accuracy (%) and ADE20k semantic segmentation mIoU for the DeiT-L backbone. We report mean $\pm$ standard error across 3 seeds.

METHOD	IMAGENET-1K ACC. (%)	ADE20K mIoU (%)
APE	83.22 $\pm$ 0.05	46.86 $\pm$ 0.06
AXIAL ROPE	83.61 $\pm$ 0.04	47.92 $\pm$ 0.06
ROPE-MIXED	83.53 $\pm$ 0.03	48.55 $\pm$ 0.06
SPIRAL ROPE (OURS)	83.97 $\pm$ 0.03	49.16 $\pm$ 0.04

B. Implementation Details

B.1. Synthetic Spatial Reasoning Tasks

Dataset construction. Both synthetic datasets consist of 224×224 RGB images on a neutral grey background (RGB 128, 128, 128), populated with non-overlapping coloured square patches of 24×24 pixels. Each image contains a red *reference* patch and a blue *target* patch; their pixel positions are sampled so that the Euclidean distance between patch centres falls within the chosen distance bin and the angular offset matches the chosen direction. Distractor patches (when used) are drawn from a fixed palette of 10 saturated colours disjoint from red/blue. Each split is generated with a fixed seed (train seed 42, test seed 123) and saved to disk in ImageFolder format for reproducibility.

- **Relative direction task.** The relative offset of the blue patch with respect to the red patch is quantised into one of 8 angular bins (E, NE, N, NW, W, SW, S, SE). To prevent the model from solving the task via simple local feature matching, we add six random-colour distractor patches per image. The train split contains 50K images uniformly distributed over the 8 classes, and the test split contains 10K images.
- **Distance task with directional shift.** The Euclidean distance between the two patches is binned into 4 buckets (Near, Mid-near, Mid-far, Far). The training split contains only axis-aligned placements (E, N, W, S), uniformly distributed over the 4 distance classes (50K images), while the test split contains both axis-aligned and diagonal (NE, NW, SW, SE) placements (10K images) sampled at the same distance bins. No distractors are used in this task.

Training. We train a DeiT-Tiny backbone (ViT patch size 16) from scratch on each task using AdamW with cosine learning-rate decay, batch size 256, minimum learning rate 10^{-6} , drop-path rate 0.1, and no data augmentation (mixup, cutmix and random erasing are disabled, since spatial augmentation would invalidate the spatial labels). Following the heterogeneous difficulty of the two tasks, we use task-specific schedules: 10 epochs with 5 warmup epochs, learning rate 5×10^{-4} , weight decay 0.05 and label smoothing 0.0 for the direction task; and 30 epochs with 10 warmup epochs, learning rate 10^{-3} , weight decay 0.1 and label smoothing 0.1 for the distance task. Model EMA is disabled. For Spiral RoPE we use $K = 16$ directions and base frequency scaling factor 1.0 (which we did not tune); the Axial RoPE baseline uses the same DeiT-Tiny backbone and the same hyperparameters but with the standard axial 2D RoPE in place of Spiral RoPE. Each task takes under 1 GPU-hour on a single A100.

B.2. Image Classification

Pre-training. We train DeiT models on ImageNet-1K following an improved training recipe. For DeiT-Base and DeiT-Small, we train for 300 epochs with a batch size of 4096, learning rate of 2×10^{-4} , weight decay of 0.3, and drop path rate of 0.1. For DeiT-Large, we train for 200 epochs with the same hyperparameters except drop path rate of 0.2. We use

AdamW optimizer with $\beta_1 = 0.95$ and cosine learning rate schedule with 20 warmup epochs. We employ the same data augmentation pipeline during training as the original DeiT implementation.

Fine-tuning. For fine-tuning at 224×224 , we train for 20 epochs with learning rate 10^{-5} , weight decay 0.1, and the same drop path rate as pre-training. For higher resolutions (384×384 and 448×448), we fine-tune from the 224×224 checkpoint for 20 epochs with the same hyperparameters. We use full crop (crop ratio 1.0) during evaluation.

Spiral RoPE Configuration. For all classification experiments, we use $K = 16$ directions with base frequency scaling factor 1.5 (i.e., $\theta_{\text{base}} = 15000$). Spiral RoPE is combined with the original absolute positional embedding (APE).

Table 9. Pre-training hyperparameters for ImageNet-1K classification.

Hyperparameter	DeiT-Base and DeiT-Small	DeiT-Large
Epochs	300	200
Batch size	4096	4096
Optimizer	AdamW	AdamW
Learning rate	2×10^{-4}	2×10^{-4}
Weight decay	0.3	0.3
β_1, β_2	0.95, 0.999	0.95, 0.999
LR schedule	Cosine	Cosine
Warmup epochs	20	20
Drop path	0.1	0.2
Label smoothing	0.1	0.1
<i>Data Augmentation</i>		
RandAugment	(9, 0.5)	(9, 0.5)
Mixup α	0.8	0.8
CutMix α	1.0	1.0
Random erasing	0.25	0.25
<i>Spiral RoPE Parameters</i>		
Number of directions (K)	16	16
Base frequency scale	1.5	1.5

B.3. Semantic Segmentation

We use UperNet as the segmentation framework with DeiT backbones initialized from ImageNet pre-trained weights at 224×224 resolution. Models are trained on ADE20K at 512×512 resolution for 160K iterations with a total batch size of 16 (8 GPUs $\times$ 2 samples per GPU). We use AdamW optimizer with learning rate 6×10^{-5} , weight decay 0.01, and polynomial learning rate decay with 1.5K warmup iterations. Weight decay is disabled for position embeddings, class tokens, normalization layers, and layer scale parameters. For Spiral RoPE models, we use the same $K = 16$ and base frequency scale 1.5 as in classification.

B.4. Image Generation

We follow the training recipe of the original DiT implementation (Peebles & Xie, 2023). Specifically, we train DiT models on ImageNet 256×256 for class-conditional image generation. All models are trained for 400K steps with a global batch size of 256. We use the AdamW optimizer with a constant learning rate of 10^{-4} and no weight decay. We employ the EMA of model weights with decay 0.9999 for evaluation. For sampling, we use DDPM with 250 steps and classifier-free guidance scale of 1.0. FID is computed on 50K generated samples using the ADM evaluation toolkit (Dhariwal & Nichol, 2021).

Spiral RoPE Configuration. For DiT experiments, we use $K = 8$ for S, B, and L models, and $K = 6$ for XL models. The base frequency scale is set to 1.5 for all models.

Table 10. DiT training hyperparameters.

Hyperparameter	Value
Training steps	400K
Global batch size	256
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.0
EMA decay	0.9999
<i>Sampling</i>	
Sampling steps	250
CFG scale	1.0
<i>Spiral RoPE Parameters</i>	
K (S/B/L)	8
K (XL)	6
Base frequency scale	1.5

C. Additional Generated Samples

We provide additional uncurated samples generated by our DiT-XL/2 model with Spiral RoPE, trained for 7 million steps on ImageNet 256×256 . For each class, we show 40 randomly generated samples. All samples are generated using classifier-free guidance with scale 4.0.



Figure 10. Class 207: Golden Retriever



Figure 11. Class 281: Tabby Cat

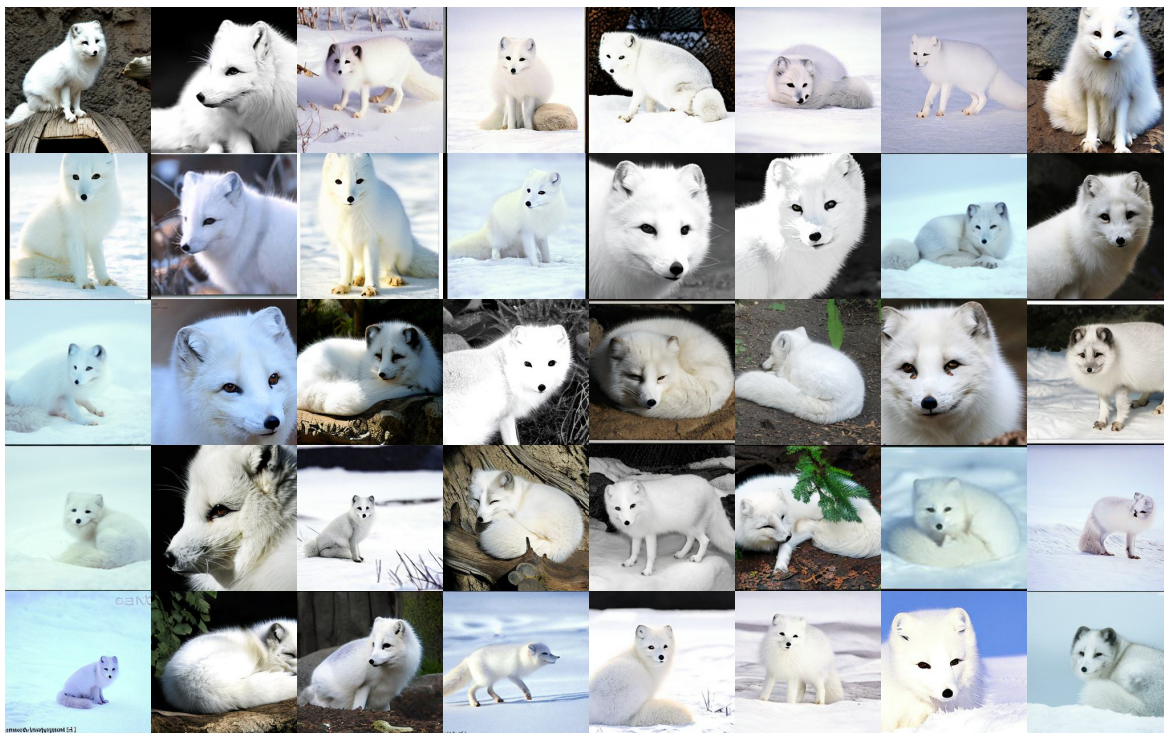


Figure 12. Class 279: Arctic Fox



Figure 13. Class 360: Otter



Figure 14. Class 33: Loggerhead Turtle



Figure 15. Class 88: Macaw



Figure 16. Class 323: Monarch Butterfly



Figure 17. Class 417: Balloon



Figure 18. Class 979: Valley



Figure 19. Class 980: Volcano